

Towards a Sustainable Design and Circular Economy for Hybrid EV Batteries

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Abstract. The HELIOS European project, funded by the H2020 Programme, aims to develop lighter, modular hybrid Li-ion battery packs for EVs that combine same-size modules with high-power and high-energy cells to meet customer demands for fast charging and long-range capabilities. Nevertheless, it is important to consider the potential environmental impacts associated with the disposal of such hybrid batteries. This includes the consumption of a wider range of raw materials, the release of hazardous substances, and the emission of greenhouse gases. To address these concerns, the HELIOS project also focuses on improving the sustainability of the battery pack by creating designs for easy reuse in second-life applications and by developing recycling strategies at the End-of-Life (EoL) of the battery pack. The project has progressed in three key research areas: 1) Designing the battery pack to incorporate possible second-life scenarios using a modular configuration and enhanced dismantling techniques. 2) Froth flotation experiments were carried out to study whether graphite could be recovered in a reusable form and to create data for future LCA calculations. 3) Parallel to the froth flotation experiments, LCA methodology supported by the ISO 14040 series was used to create a baseline to evaluate the environmental impacts of the hydrometallurgical recycling of the NMC811-graphite cells. This baseline could be used later to assess the impact of graphite recovery by froth flotation, as the current LCA baseline excludes the recovery of anode materials (graphite and LTO).

Keywords: Hybrid EV Battery · Recycling · 2nd Life · Dismantling · Refurbishment · Life Cycle Assessment · Environmental Impacts

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1 Introduction

The growing electric vehicle (EV) markets have increased the demand for raw materials needed in the production of lithium-ion batteries (LIBs) to power the transition towards transportation sector electrification [1]. Furthermore, this fast-growing rate of raw material consumption also raises concerns regarding material availability, for which recycling offers a potential solution to narrow the gap between the supply and demand of lithium-ion battery (LIBs) raw materials. Consequently, the European Union has imposed new stricter recycling requirements for critical materials and from 2030 onwards, 70% of LIBs materials need to be recovered to reduce impacts due to mining [2]. However, the efficient separation of the active components and their recovery from battery wastes remains a challenge.

In addition to the regulations encompassing a wide array of requirements designed to encourage the recycling of key battery raw materials [3, 4], the European Union has introduced new targets aimed at promoting sustainability [5]. These include carbon footprint performance class requirements, applicable from 1 July 2025, for electric vehicle batteries and the reduction of battery weights [6]. The new EU laws underscore the importance of designing batteries for easy reuse in second-life applications, thereby paving the way for a more sustainable future in energy storage. Compared to directly recycling lithium-ion batteries after their electric vehicle use, the carbon footprint and energy use of batteries recycled after their second life can be reduced by 8 to 17% and 2 to 6%, respectively.

Utilizing Second-Life batteries can provide numerous advantages that contribute to their overall sustainability. Firstly, they extend the useful life of EV batteries, delaying their disposal and reducing waste [7]. Moreover, they can offset the need for new battery production and its associated environmental impact [8]. Finally, they can provide affordable energy storage solutions that support the integration of renewable energy into the grid [9]. At the European level, the potential for Second-Life batteries is significant. With millions of EVs expected to be on the roads in the coming years, a vast number of batteries will eventually reach their End-of-Life (EoL) [10]. These batteries often retain a substantial portion of their original capacity and can still provide valuable service in second-life applications prior to the final EoL recycling [11].

Currently, lithium-ion batteries are constantly undergoing development to improve EVs performance requirements for higher power density, fast charging, and longer cycle life. Therefore, a large variety of cell chemistries as well as pack, module and cell designs have entered the market to meet these demands [12]. Nonetheless, designs that are being developed to offer improved LIBs performance and longer lifetimes often result in new material selections and combinations that may not have established recycling strategies. For example, material selection can include raw materials with low economic value, which can create challenges for their separation within existing metallurgical recycling infrastructures. The wide variety of cell chemistries and module and cell designs in the market means that pyrometallurgical treatment is currently the only economically viable option, as it offers enough flexibility in terms of process feed, although with the caveat that materials such as graphite or Li are lost. [13–15].

The HELIOS project, funded by the European Union's Horizon 2020 program, has developed a series of modular and lightweight hybrid Li-ion battery packs. These packs

combine high-power cells comprised of NMC811 ($LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$) cathode and a $Li_4Ti_5O_{12}$ (LTO) anode, with high-energy cells featuring an NMC811 cathode and graphite anode. The aim of the project is to explore suitable solutions for recycling these battery modules at their EoL, as well as dismantling and refurbishment approaches to promote second-life usage and recycling. The outcomes of these efforts will be integrated into Life Cycle Assessment (LCA) models to support a sustainable and circular design for hybrid batteries.

2 Methodology

Three approaches are outlined in this study following strategies that are being developed in HELIOS project to improve the sustainability of the hybrid battery pack:

- Novel manufacturing and dismantling processes enable second life use and new methodologies for recycling and metal recovery.
- Graphite recovery from mixed NMC811-graphite-LTO chemistries is studied experimentally by froth flotation. The experiments simulate the behaviour of a perfectly liberated model NMC811-graphite-LTO system in the froth flotation process without the effect of binder, electrolyte, solid electrolyte interphase (SEI), current collector pieces and plastics to test froth flotation's potential for inclusion as part of battery recycling strategies.
- The environmental impacts of recovering Co, Ni, Mn, and Li from NMC811-graphite cells by hydrometallurgical processes were simulated and an LCA was performed to obtain an overview of the environmental impact of recovering these elements.

The outcomes of these three approaches are planned to be included in more detailed cradle-to-grave LCA models. Furthermore, an additional LCA focused on a process to recover the nickel by using a hydrometallurgy process is also to be undertaken.

3 Novel Manufacturing and Dismantling Processes for Hybrid Battery Packs

This section explains the methodological strategies to enhance the efficiency and safety of recycling processes, with an emphasis on circular design principles. The principles of the circular economy revolve around designing out waste and pollution whilst maximizing the useful life of products and materials. The main principles where the HELIOS actions are focused are as follows:

- Design Out Waste and Pollution [16]: In a circular economy, resources are used efficiently, and waste is designed out of the system from the beginning. This means that products are designed and optimized for a cycle of disassembly and reuse, which can significantly reduce waste and pollution.
- Keep Products and Materials in Use [16]: The focus is on creating a "closed loop" system where resources are not just consumed and discarded but are reused, repaired, or refurbished to extend their life. This includes considering second-life applications for products or components that can no longer serve their primary purpose.

- Rethinking of Business Models [16]: Companies are encouraged to rethink their business models to support longer product lifetimes, modular design, product sharing, leasing or take-back schemes that can help keep products and materials in use for longer. By aiming to improve the efficiency of each stage, new methods to design products via a holistic approach are applied, taking into consideration their entire lifecycle, from disassembly to manufacturing and (re)assembly - the main methods are described as follows:

The new procedures are based on a thorough Analysis of Existing Techniques used in the industry that include:

Robotic Battery Disassembly: currently battery packs are manually dismantled, which requires skilled labour and specialized tools. The introduction of robotic disassembly based on current approaches will improve the performance of the process, reduce the risk of harm to workers and decrease costs, making recycling more economically viable. Another advantage is that robots can perform tasks with extremely high precision, which will reduce the risk of damaging valuable components and will ensure that valuable battery cells are handled carefully. During the disassembly process valuable data about aging, and reliability. Etc. could be collected and analyzed.

Process Optimization: development of new procedures for dismantling and refurbishment to streamline and automate the recycling process. by involving deconstruction planning where CAD design is used for pre-planning the disassembly process where the robot follows a predefined plan to remove components in a specific order. This approach includes also exploiting the modular design where the process is simplified because robots remove not only components but entire modules. Another advancement in process optimization is that the design is done in such a way that at almost all steps disassembled components are designed to be manufactured from the same material. This will facilitate second life use and introduce new methodologies for recycling and metal recovery.

The HELIOS battery pack prototypes represent a significant advancement in battery technology, a unique combination of performance, flexibility, and cost-effectiveness through the combination of high-energy (HE) cells for increased range and high-power (HP) cells for faster charging in separate modules. By utilizing a modular design with the same size and shape plastic housings or "egg-box" (each cell inside is separated like eggs in a cardboard box), these battery packs provide a versatile solution that can be seamlessly integrated into different EV models to accommodate various power and energy demands (Figs. 1 and 2).

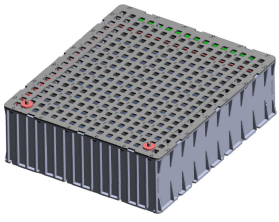


Fig. 1. HE module in "egg-box"

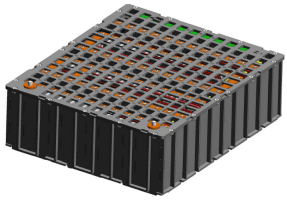


Fig. 2. HP module in "egg-box"

Either four HE or HP modules are then packaged into the so-called "subpack" that contains its own cooling circuit (Fig. 3).

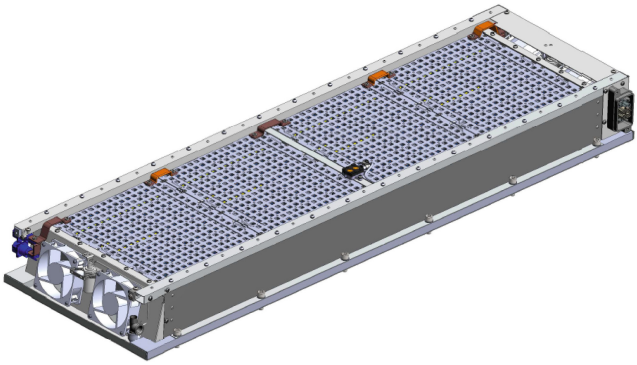


Fig. 3. Helios subpack with 4 identical modules inside

According to the needs of the original equipment manufacturer (OEM) and end-users, these subpacks can be combined in any way to meet the demands of a specific application, hence they are customizable to drivers' requirements and scale-up for different size of EVs. The interconnections between the subpacks inside a full HELIOS battery pack are quite simple and based on high voltage (HV) copper busbars and wires, which allows easy removal of individual subpacks for reuse in second-life applications like energy storage.

Furthermore, at the ultimate EoL, the complete dismantling of a subpack for the final recycling of all materials inside the "egg-box" modules is planned to be a straightforward and mainly automated process.

4 Froth Flotation as a Potential Strategy for Graphite Recovery from NMC811-Graphite-LTO Chemistries

The recycling of LIBs is primarily linked to the economic worth of the metals they hold. At present, only the most valuable metals, including Cobalt, Nickel, and Copper, are commonly extracted. [15]. From the HELIOS battery cells, Ni and Cu are predicted to face future supply risks [17] whereas Co, natural graphite and – more recently - Li, Ti, Mn and Al are also categorized as critical raw materials by the EU [2] highlighting the importance of developing strategies for the optimal recycling of these elements.

The material selection of the HELIOS battery pack poses challenges for critical raw materials recovery based on the industrial-scale extractive options currently employed for recycling spent LIBs. Pyrometallurgical processes, widely used for their high reaction rates and lack of requirement for passivation or dismantling, produce a metal alloy and a slag that contains the various oxides from Li-ion battery modules. However, this approach significantly loses critical battery materials and graphite, which is reduced to CO_2. [12, 14, 18]. Furthermore, the loss of these components will make it difficult to reach the future EU battery recycling requirements.

Consequently, research has shifted more towards hydrometallurgical recycling as a larger number of elements can be recovered, often with high selectivity [19]. Traditional hydrometallurgical recycling uses acidic media to leach metals from waste Li-ion battery electrodes, producing a solution for purification and recovery through precipitation and solvent extraction. These processes can be complex, yielding high-purity metals, but also significant secondary pollution [20, 21]. In addition, hydrometallurgical recycling is not an optimal method to recover graphite spent LIB mixtures. Graphite is porous and adds volume to the feed material, increasing water and acid consumption and producing a viscous slurry that is difficult to filter and wash [22]. Graphite particles may also get damaged during leaching, preventing its reuse it as an anode material [23, 24].

Although graphite recovery has not been the focus of most recycling processes so far, anodic graphite represents 14 – 22% of a LIBs weight and can help to satisfy the future EU recycling requirements [17]. Thus, froth flotation has recently attracted the attention for use in graphite recycling, since it is a physical separation method in which the graphite structure can be maintained, thereby allowing the direct reuse in new batteries manufacturing [17, 23, 25, 26]. Froth floatation is able to separate lithium metal oxide powder from the graphite in Li-ion batteries, by exploitation of their different wettability characteristics. Bubbles introduced into a stirred vessel containing the black mass collect hydrophobic materials, producing a graphite-rich froth. Additional reagents can enhance froth stability and promote wettability differences [27]. Direct recycling of graphite could be economically attractive if its structure were preserved and if it could be recovered with battery grade purity (> 99%). [26].

Material and methods, discussion, and preliminary results by application of this novel method in the HELIOS context are described in the following sections.

4.1 Materials and Methods

Pure battery-grade graphite, NMC811 and LTO particles were mixed according to the expected active material composition of the HELIOS battery modules, i.e., 50% of NMC811—as cathodes in both cell types contain this active material—25% of LTO and 25% of spherodized graphite particles. The particle size distributions of these model materials (shown in Fig. 4) were measured with a laser diffraction particle size analyser (Malvern Panalytical Mastersizer 3000, Malvern, UK) and the sources of chemicals used are listed in Table 1.

A laboratory-scale, 1 L flotation cell (Lab Cell-60mm FloatForce mechanism, Outotec, Espoo, Finland) was used for the flotation experiments. 40 g of the active particle mixture (50% NMC811; 25% graphite; 25% LTO) was added to 1 L of ion exchanged water and mixed by impeller at 1000 rpm for 3 min, after which the rotation speed was reduced to 850 rpm and 8 ppm of MIBC frother was added to the cell. The pulp was conditioned with MIBC for 2 min, and then air was introduced to the cell through the impeller at a flowrate of 2 l/min, to create bubbles and initiate the flotation experiment. Collection of the froth started when the froth rose to a suitable level and it was then sampled at time intervals of 0–1 min, 1–3 min, and 3–8-min to study the flotation process kinetics. The collected concentrate samples and the remaining pulp in the flotation cell were filtered and dried in a convection oven.

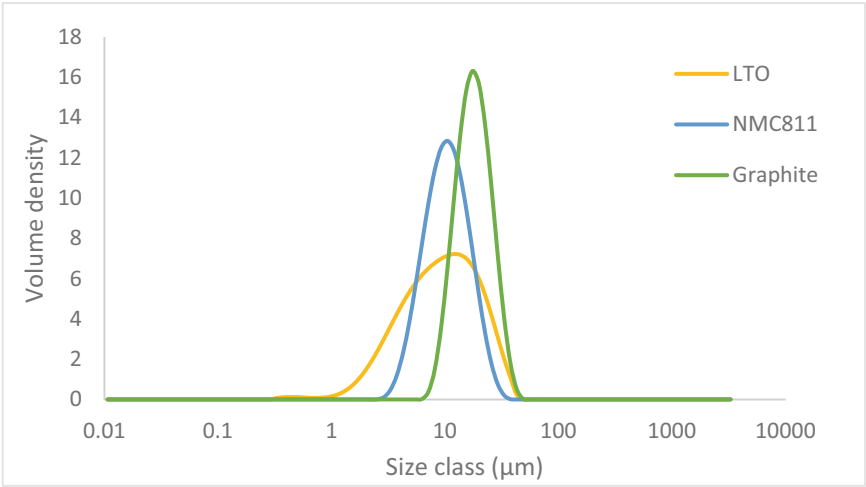


Fig. 4. Particle size distributions of model black mass.

Table 1. Materials and reagents used in froth flotation experiments of model black mass.

Distributor	Chemical	Purity [%]
MSE Supplies, Tuscon, AZ, USA	$LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$	Battery grade
MSE Supplies, Tuscon, AZ, USA	$Li_4Ti_5O_{12}$	Battery grade
ProGraphite GmbH, Untergriesbach, Germany	Spherodized graphite	99.95
Sigma-Aldrich, St. Louis, MO, USA	Methyl isobutyl carbinol	98

These dried samples were subsequently weighed, and their composition analysed by XRF (Oxford Instruments, X-MET 5000, Abingdon, UK) and SEM-EDS (Scanning Electron Microscopy - Energy Dispersive X-ray Spectroscopy) (MIRA3 SEM, Tescan, Brno, Czech Republic and UltraDry Silicon Drift Energy Dispersive X-Ray Spectrometer combined with NSS Microanalysis Software (Thermo Fisher Scientific, Waltham, MA, USA)). A schematic representation of the froth flotation experimental procedure is presented in Fig. 5.

4.2 Findings and Analysis

The hydrophobic graphite particles are collected in the froth phase, while the hydrophilic lithium metal oxides remain in the tailings. Nevertheless, with increased flotation time, the grade of the recovered graphite was found to decline - as shown in the SEM images presented in Fig. 6.

Rinne et al. [28] and Vanderbruggen et al. [17] have attributed this lowering of graphite grades to entrainment, which occurs as lithium metal oxide particles are carried with process water to the froth layer. Entrainment is an unselective recovery mechanism

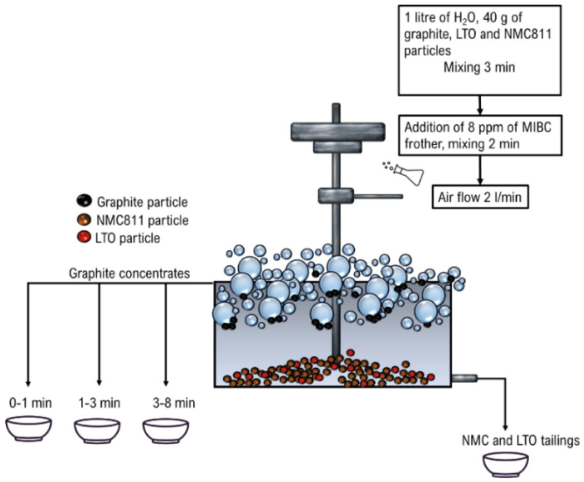


Fig. 5. Flowsheet of the froth flotation experiments.

and fine particles are more prone to be recovered by this mechanism since their settling rate is slower cf. Large particles [27]. Vanderbruggen et al. [17] measured a linear correlation between NMC111 contamination in the recovered graphite and water pulled from the flotation cell, indicating that graphite grade declines due to entrainment of particles below 10–15 μm size range. Interestingly, Vanderbruggen et al. [17] also measured significant amounts of pure NMC111 particles attaching to air bubbles in their captive bubble experiments. It is therefore possible that NMC811 and LTO particles exhibit both entrainment and true flotation behavior. The improvement of graphite grade therefore depends on the prevention of lithium metal oxide recovery by any of these mechanisms. For example, Rinne et al. [26] improved the grades of recovered graphite by increasing LCO particle sizes with selective flocculation.

Use of selective depressants targeting the LTO and NMC811 particles could also improve the graphite grades by reducing true flotation [17]. Lithium is known to leach to the process water and Li + ions have been shown to stabilize froths, thereby increasing water recovery and entrainment. Therefore, washing the black mass before flotation has resulted in improved graphite grades. [26, 28] Other options for reducing entrainment include lowering solid-liquid ratios [29] and selectively flocculating lithium metal oxides into larger particle clusters [26].

Figure 7 shows the cumulative grade vs. cumulative recovery curves and flotation kinetics of the 50% NMC811–25% LTO-25% graphite mixture. The data points represent the arithmetic mean from four repeated experiments under identical conditions and the standard deviations of the results. The large experimental variation can be partly attributed to the manual froth collection. On average 87.8% of the graphite could be recovered with 94.8% grade and 50% of the graphite can be recovered with 98% grade. These results show that froth flotation has the potential to be used for the direct recycling of LIBs, although further research to optimize the froth flotation process is needed to achieve consistently > 99% grades.

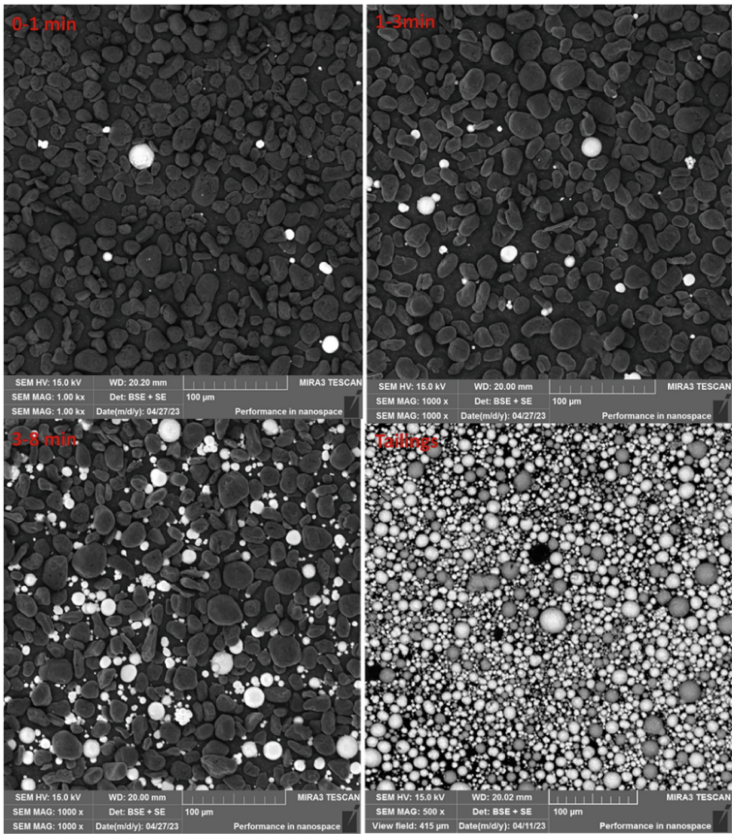


Fig. 6. Comparison of the graphite concentrates collected during 0–1 min, 1–3 min and 3–8 min flotation times and the remaining tailings. Black particles are graphite, grey particles are LTO, and white particles are NMC811.

The kinetics of graphite flotation were determined by fitting the graphite recoveries collected at different time intervals to the classical first order kinetic model according to Eq. (1) [27, 28] to obtain the rate constant k = 0.85 for the electrode mixture studied.

$$R(t) = R_{max}\left(1 - e^{-kt}\right) \tag{1}$$

where, R(t) is the recovery of graphite at time (t), Rmax is the maximum recovery after the experiment and k is the rate constant.

Such kinetic data, graphite grades information and recoveries could be utilized in the HSC simulations and LCA calculations related to the recycling of HELIOS modules.

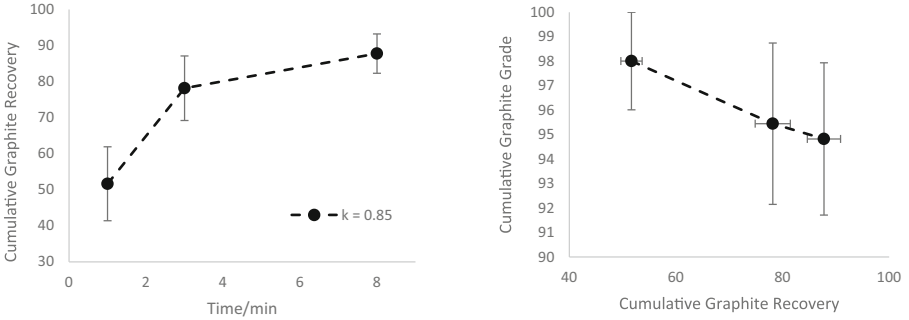


Fig. 7. Cumulative grade of graphite as a function of graphite recovery and graphite recovery as a function of flotation time during froth flotation of model black mass composed of 50% NMC811, 25% graphite and 25% LTO. The curves represent repeated experiments under identical conditions.

5 Environmental Impacts of Recovering Co, Ni, Mn, and Li from NMC811-Graphite Cells by Hydrometallurgical Processes: A Simulation and LCA Study

Process simulation is a useful tool when reliable industrial data is unable be obtained. In this case, the HSC Chemistry® Sim software module (METSO, Pori, Finland) was utilized to obtain the life cycle inventory (LCI) of the emerging hydrometallurgy recycling process. An extensive literature review was carried out to obtain the parameters for the process simulation and chemical reactions. Open LCA software by Green Delta was utilized to generate the life cycle analysis with the support of the Ecoivent 3.6 database developed by the Swiss Centre for Life Cycle Inventories.

5.1 Goal and Scope

The goal of the study was to understand the environmental impacts of recycling black mass with a high content of nickel. The functional unit selected is 1 kg of recycled black mass. For the environmental impacts it was planned that the LCA-related work undertaken as part of HELIOS would not only cover the carbon footprint of the EV batteries, but also consider additional impact categories of relevance for the actors involved in their mining, manufacture, and use by a grave to gate study. Preliminary treatment of the black mass by crushing, magnetic separation, and sieving is not included in the scope of this study along with the binder and electrolyte effects on the system.

5.2 System Boundary

The system under study is shown in Fig. 8 and depicts the hydrometallurgy process for battery recycling proposed for the black mass with high nickel content. The study boundary is composed of leaching, copper cementation, iron and aluminium removal by hydrolysis, manganese solvent extraction and precipitation, cobalt and nickel solvent extraction and crystallization, and lithium precipitation. The effluents from the system are

collected and treated via a neutralization step. Nonetheless, the environmental impacts associated with waste management and further wastewater treatment associated with the overall process were beyond the scope of this work.

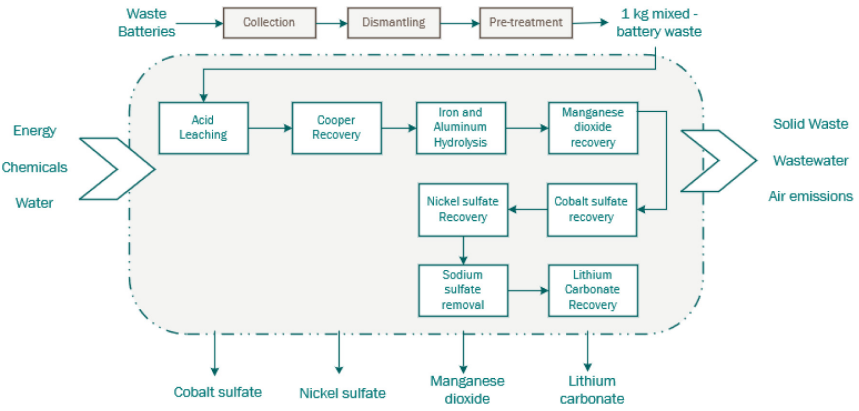


Fig. 8. System boundary and flowsheet diagram of the battery recycling process to study.

5.3 Impact Assessment Methods and Indicators

An LCA was carried out using Open LCA software, and the Ecoinvent database 3.6, with three different impact methods selected to analyse the impact categories. Under this context and aligned with the recommendations in ISO 14044 [29], the Environmental Product Declaration, EPD® system [30] and the PEF method [31], the impact categories shown in Table 2. Are suggested to study the environmental behaviour of the battery solutions. Table describes the environmental category analysed the method used and its description.

5.4 Process Descriptions

The black mass underwent preliminary treatment involving crushing, magnetic separation, and sieving, to achieve a particle size smaller than 0.125 mm. The material that was taken as a reference was investigated previously by Roy et al. (2019) who studied NMC 111 [32]. Nonetheless, the mass percentage of the LCO, LNO, and LMO were modified to stoichiometrically increase the LNO content to match the NMC811 composition. The chemical composition of the black mass fed to the system is outlined in Table (Table 3).

The black mass was fed to a leaching system that comprised of 2M H2SO4 and 4% of H2O2. The pregnant leaching solution (PLS) was further treated in a complex battery recycling circuit to recover copper as metallic copper powder through cementation, manganese recovery as oxide (MnO2), through solvent extraction and oxidative precipitation, recovery of cobalt and nickel respectively as nickel sulphate (NiSO4·6H2O), and cobalt sulphate (CoSO4·7H2O) through solvent extraction and crystallization, and

Table 2. Impact categories analysed, the method used, and units.

Abbreviation	Impact category	Unit (Method)
GWP	Global Warming Potential	kg CO_2-eq (CML)
ADP	Abiotic depletion (elements)	kg Sb-eq (CML)
ADPf	Abiotic depletion (fossil)	MJ (CML)
AP	Acidification potential	kg SO_2-eq (CML)
EP	Eutrophication potential	kg PO_4 3—eq (CML)
FW-AE	Fresh water aquatic ecotoxicity	kg-DCB-eq (CML)
MA-EX	Marine aquatic ecotoxicity	kg-DCB-eq (CML
ODP	Ozone depletion	kg CFC-11-eq (CML)
POCP	Photochemical ozone creation potential	kg ethene-eq (CML)
T-EX	Terrestrial ecotoxicity	kg-DCB-eq (CML)
FPM	Particulate matter	kg PM2.5 eq (ReCiPe)
LU	Land use	m^2 x yr annual crop land (Recipe)
HT-C	Human toxicity potential – Cancer	CTUh (USEtox 2.01)
HT-NC	Human toxicity potential – Non-Cancer	CTUh (USEtox 2.01)

Table 3. Chemical composition of the black mass fed to the system.

Component	$LiNiO_2$	$LiCoO_2$	$LiMnO_2$	Cu	Al	Fe	C	Total
wt%	54.66%	6.57%	6.85%	3.56%	0.78%	1.23%	26.35%	100%

lithium carbonate (Li2CO3). After every stage, all effluents collected and subject to a simple neutralization step were metals (except lithium, calcium, and sodium) which were removed at pH 7. Additionally, the flowsheet removes Glauber salt by increasing the concentration of Li through evaporation to remove (Na2SO4·10H2O) through crystallization, whereas aluminium and iron are taken out of the system as oxides via hydrolyzation (Table 4).

5.5 Process Performance

The selected mass flow for the study was 7000-ton of black mass/year as Rinne et al. (2021) previously proposed in their study where a LIBs and NIMH battery recycling system was investigated [21]. The flowsheet simulation was successfully recovered. ∼92% Li, 95% Co, 94% Ni, and 94% Mn. The estimated purity of the recovered materials is shown in Table 5.

Table 4. Simulation parameters for flowsheet simulation in HSC-Sim

Leaching	Unit	Value	Reference
H_2O_2	%	5	[33] [5]
H_2SO_4 concentration	gr/L	196	[33]
S/L ratio	g/L	100	[34]
Mn solvent extraction	Unit	Value	Reference
pH	-	4	[35]
O/A ratio in extraction	vol/vol	1:1	[35]
O/A in stripping	vol/vol	8:1	[35]
H_2SO_4 concentration in stripping	mol/L	0.5	[35]
$MnSO_4$ concentration in scrubbing	g/L	4	[35]
O/A ratio in scrubbing	vol/vol	10	[35]
Mn precipitation	Unit	Value	Reference
SO_2 partial pressure in SO_2/O_2	p.%	6	[36]
pH	-	3–4	[36]
O_2 utilization efficiency	%	33	Estimate
Al and Fe removal	Unit	Value	Reference
pH in neutralization	-	4	[20]
Co solvent extraction	Unit	Value	Reference
pH	-	5.4	[20]
Co loading in extractant	g/L	~17	[37]
O/A in extraction	vol/vol	1:1	[20]
O/A in stripping	vol/vol	1:4	Estimated
H_2SO_4 concentration in stripping	g/L	10	[20]
O/A scrubbing	vol/vol	10:1	Estimated
$CoSO_4$ concentration in stripping	g/L	10	[20]
Ni solvent extraction	Unit	Value	Reference
pH	-	6.5	[20]
O:A in extraction	vol/vol	1:2	Estimate
O:A in stripping	vol/vol	4:1	Estimate
H_2SO_4 concentration in stripping	g/L	10	[20]
O:A in scrubbing	vol/vol	10	[20]
$NiSO_4$ concentration in scrubbing	mol/L	0.3	[20]
Sodium sulphate crystallization	Unit	Value	Reference

(*continued*)

Table 4. (*continued*)

Leaching	Unit	Value	Reference
Li concentration	g/L	20 g/L	[38]
Na_2SO_4 solubility	g/L	12	[38]
Lithium carbonate crystallization	Unit	Value	Reference
Li concentration	g/L	1	[38]
pH	-	12	[38]

Table 5. Material recovered from the software simulations and their purities (%)

Material	Purity	Unit
MnO_2	98.41	%
Li_2CO_3	97.79	%
$NiSO_4$	99.61	%
$CoSO_4$	99.34	%

5.6 Life Cycle Inventory

A comprehensive life cycle inventory is presented in Table 6. It must be noted that even though life cycle assessment using Ecoinvent 3.6 is an optimal tool to understand the environmental impacts of a process when industrial data is not available it has its limitations. For example, solvent extraction chemicals (D2EHPA, Cyanex 272, TBP) were not included in the LCI, moreover, cobalt sulphate had to be replaced stoichiometrically with metallic cooper and oxygen gas was excluded from the LCI due to lack of data availability in the Ecoinvent data 3.6. Due to these reasons, no results related to the excluded materials were generated to analyse a more comprehensive life cycle assessment, nevertheless, this investigation was conducted using the most up-to-date information accessible during the period of this study.

5.7 Impact Analysis

The total environmental impacts are summarized in Table 7, while the contribution of every material to each category is depicted in Fig. 9. It is interesting to observe that through the impact categories nickel sulphate, sodium hydroxide, and sulfuric acid had the highest contribution to the environmental impacts. Nickel sulphate is used in minimum amounts for the purification of the recovered nickel while sodium hydroxide and sulfuric acid are used in several different steps of the flowsheet.

For GWP, the more important contributors were associated to the use of sodium hydroxide (35%), nickel sulphate(20%), hydroxide peroxide and the generation of steam (∼11%). ADP contributions were nickel sulphate 50% and sodium hydroxide and sulfuric acid with ∼19%. For ADPf contributions were related to the use of sulfuric acid

Table 6. Life cycle Inventory (LCI)

Input			Output		
Leaching					
Black mass	1.00	kg/kg	Leaching Residue	0.22	kg/kg
Sulfuric acid	1.49	kg/kg	$O_{2(g)}$	0.12	Nm^3/kg
Hydrogen peroxide	0.29	kg/kg	$H_{2(g)}$	0.01	Nm^3/kg
			H_2O(g)	2.86	Nm^3/kg
Cu Cementation					
Iron scrap	0.02	kg/kg	Cu	0.03	kg/kg
Iron-Aluminum removal					
$O_{2(g)}$	0.01	Kg/kg	O2(g)	0.01	Nm^3/h
NaOH	0.56	kg/kg	$Fe(OH)_3$	0.06	kg/kg
			$Al(OH)_3$	0.02	kg/kg
Mn Recovery					
Manganese sulphate	0.01	kg/kg	MnO_2	0.07	kg/kg
Sulfuric acid	0.10	kg/kg	$O_{2(g)}$	0.01	Nm^3/h
NaOH	0.14	kg/kg			
$O_{2(g)}$	0.03	kg/kg			
$SO_{2(g)}$	0.00	kg/kg			
Co Recovery					
Cobalt Sulphate	0.01	kg/kg	Cobalt Sulphate	0.15	kg/kg
Sulfuric acid	0.11	kg/kg			
NaOH	0.09	kg/kg			
Ni Recovery					
Ni	0.12	Kg/kg	Nickel sulphate	1.25	kg/kg
Sulfuric acid	0.76	kg/kg			
NaOH	0.62	kg/kg			
Lithium Recovery and Na_2SO_4 Removal					
Na_2CO_3	0.27	kg/kg	Lithium carbonate	0.19	kg/kg
NaOH	0.00	kg/kg	Sodium sulphate(anhydrite)	3.16	kg/kg
Effluent Treatment					
Lime	0.01	kg/kg	Wastewater	8.41	kg/kg
			Solid Waste	0.04	kg/kg

Table 7. Life cycle assessment impact categories with total results

Impact category	Unit	Value
GWP	kg CO_2 eq	3.66
ADP	kg Sb eq	2.14×10^{-4}
ADPf	MJ	50.40
AP	kg SO_2 eq	0.19
EP	kg PO_4-eq	1.15×10^{-2}
FW-AE	kg 1.4-DB eq	15.80
MA-EX	kg 1.4-DB eq	17458.86
ODP	kg CFC-11 eq	1.39×10^{-6}
POCP	kg C_2H_4 eq	7.69×10^{-3}
T-EX	kg 1.4-DB eq	1.78×10^{-2}
FPM	kg $PM_{2.5}$ eq	4.91×10^{-2}
LU	m^2a crop eq	3.04
HT-C	cases	3.43×10^{-10}
HT-NC	cases	2.72×10^{-10}

(21%), sodium hydroxide (28%), steam and hydrogen peroxide (∼11%). AP contributions were primarily the result of nickel sulphate usage (80%) and sulfuric acid (13%). EP contributions were related to nickel sulphate and sodium hydroxide usage (∼36%). For the FW-AE category, 81% is associated with nickel sulphate and 9% with sodium hydroxide. While, for MA-EX nickel sulphate(66%), sodium hydroxide (17%), and sulfuric acid (7%) are the primary associations. ODP contributions are associated to the usage of nickel sulphate 85%. POCP contributors were associated to nickel sulphate80%, and 13% of sulfuric acid. T-EX contributions were nickel sulphate 50%, sodium hydroxide at 25%, and sulfuric acid at 8%. FPM impacts were associated with the usage of nickel sulphate with 78% and sulfuric acid with 13%. In contrast, LU associated impacts were nickel sulphate(70%), and sodium hydroxide and cobalt production (∼11%) each.

It was observed in the results of the environmental impacts related to toxicity (HT-C and HT-NC) the main contributors were sodium hydroxide, hydrogen peroxide and nickel sulphate use throughout the categories. It must be highlighted that the units of these categories are quantified in several cases. For HT-C Sodium hydroxide 23%, hydrogen peroxide 27%, nickel sulphate 19%, steam, soda ash and sulfuric acid ∼8%. For HT-NC the biggest contributors were sodium hydroxide 32%, nickel sulphate 16% production and sulfuric acid 35%. Additionally, soda ash contributions through the categories were $> 7\%$ except for HT-C. Electricity contributions were $> 5\%$ in every category, therefore, it can be concluded that the battery recycling system is not highly energy dependent.

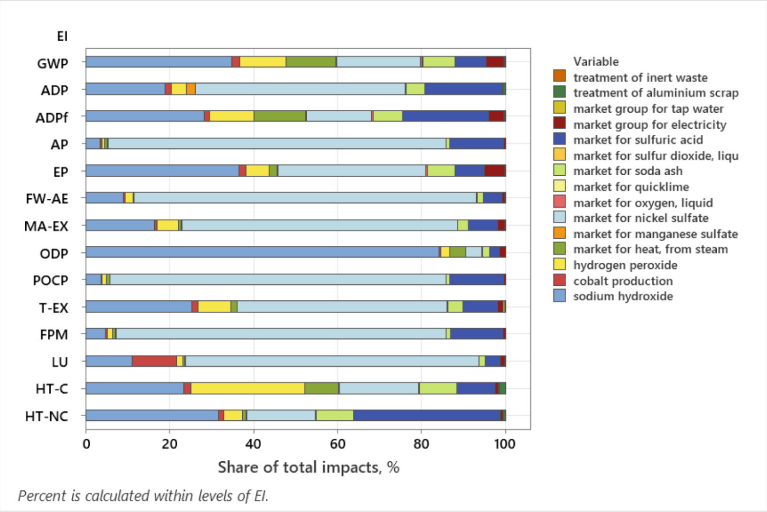


Fig. 9. Depicts the environmental impacts of every impact category assessed by Recipe hierarchy (H), CML IA baseline and USEtox

6 Conclusions

This study presents a preliminary comprehensive approach towards enhancing the efficiency of battery recycling processes within the HELIOS project. The study focuses on a set of integrated strategies that consist of three distinct actions to improve the environmental sustainability of hybrid batteries. The findings and main conclusions are outlined below:

1. The design for Disassembly/Manufacturing/Assembly focuses on creating efficient and safe methods for each stage of the battery lifecycle. This includes an analysis of existing techniques, the introduction of robotic disassembly, and the optimization of recycling processes. These methods are expected to reduce waste, lower costs, and increase worker safety.
2. Results of the froth flotation experiments demonstrate good separation between fully liberated graphite-NMC811-LTO electrode mixture. 88% of the graphite could be recovered with 94.8% purity and approximately 50% of the recovered graphite was recovered with a 98% grade. The kinetic flotation data aids in simulating the HELIOS battery pack recycling process. Removing graphite reduces environmental impact and acid use, and recovered graphite can be reused. Optimization of flotation parameters and pretreatment steps are needed to increase graphite grade. This process also enables TiO_2 recovery.
3. The results show that HSC sim process simulation baseline for recycling NMC811 cathodes provides an optimal basis for future LCA calculations for designing a recycling flowsheet for the HELIOS battery modules. It successfully recovered battery-grade lithium, cobalt, nickel, and manganese. LCA results suggest that most of the environmental impacts come from the use of nickel and sodium hydroxide. The results

from the froth flotation experiments can be incorporated into the baseline to broaden the scope of the flowsheet.

In conclusion, the integrated strategies proposed in this study have the potential to enhance the efficiency and sustainability of battery recycling processes. The findings suggest that the design for Disassembly/Manufacturing/Assembly, froth flotation experiments, and HSC sim process simulation baseline can significantly reduce waste, lower costs, and increase worker safety while recovering valuable materials from hybrid batteries while providing the basis to develop the further cradle-to-grave LCA models.

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List of abbreviations

ADP	Abiotic depletion (elements)
ADPf	Abiotic depletion (fossil)
AP	Acidification potential
EoL	End-of-Life
EP	Eutrophication potential
EV	Electric vehicle
FPM	Particulate matter
FW-AE	Fresh water aquatic ecotoxicity
GWP	Global Warming Potential
HE	High-energy
HP	High-power
HSC	H (enthalpy), S (entropy), and Cp (heat capacity)
HT-C	Human toxicity potential – Cancer
HT-NC	Human toxicity potential – Non-Cancer
HV	High voltage
LCA	Life Cycle Assessment
LIB	Lithium-ion batteries
LTO	Lithium, titanium and oxygen
LU	Land use
MA-EX	Marine aquatic ecotoxicity
NMC	Nickel, Manganese and Cobalt
ODP	Ozone depletion
OEM	Original equipment manufacturer
POCP	Photochemical ozone creation potential
SEI	Solid electrolyte interphase
SEM-EDS	Scanning Electron Microscopy - Energy Dispersive X-ray Spectroscopy
T-EX	Terrestrial ecotoxicity

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